

MACHINE LEARNING DATA ANALYST – AGRIBUSINESS

WEEK 2

EXPLORATORY DATA ANALYSIS (EDA)

Agricultural Crop Yield Analysis in Indian States

INTERNSHIP PROJECT REPORT

Prepared By

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Internship Period

04 September 2026 – 02 October 2026

Task

Week 2 – Exploratory Data Analysis

Domain

Agriculture • Machine Learning • Data Analytics

PROJECT PURPOSE

This report documents a structured exploratory analysis of agricultural crop-yield data from Indian states. The objective is to understand the distribution, variation, relationships, and regional or seasonal patterns present in the data and to convert those patterns into clear, evidence-based agricultural insights.

DATASET SNAPSHOT

Records	Features	States	Crops	Seasons
19,685	10	30	55	6

WEEK 2 SUBMISSION • PROFESSIONAL EDA REPORT

1. Executive Summary

This report presents the Exploratory Data Analysis (EDA) conducted on a cleaned agricultural crop-yield dataset covering crop production across Indian states and agricultural seasons. The final analytical dataset contains 19,685 records and 10 variables, representing 55 crops, 30 states, and 6 seasons.

The EDA follows a systematic workflow beginning with structural inspection and categorical analysis, followed by numerical distribution analysis, crop-wise and state-wise comparisons, seasonal and year-wise trend analysis, correlation analysis, outlier investigation, and multi-dimensional analysis. Visualizations are used alongside statistical summaries so that important findings can be interpreted from both numerical and graphical perspectives.

A major observation is the large variation in production and yield across crops and geographic regions. Production has the strongest positive linear relationship with Yield among the examined numerical variables, while Annual Rainfall and the other agricultural input variables show comparatively weak linear relationships with Yield.

Outlier analysis identified substantial numbers of extreme observations. These were investigated rather than automatically removed because extreme agricultural values can represent genuine differences in cultivation scale, environmental conditions, or productivity. Overall, the EDA establishes a strong analytical foundation for future feature engineering and machine learning.

2. Objectives of the Analysis

Understand the structure, scale, and quality of the cleaned agricultural dataset.

Analyze the frequency and representation of crops, states, and agricultural seasons.

Study distributions and variability of Area, Production, Rainfall, Fertilizer, Pesticide, and Yield.

Compare crop-wise, state-wise, and season-wise yield and production performance.

Identify year-wise trends and relationships that can guide future machine learning work.

Detect and document statistical outliers and investigate Crop × Season and State × Crop patterns.

3. Dataset Overview

The dataset contains crop identity, year, season, state, cultivated area, production, rainfall, fertilizer usage, pesticide usage, and yield. These variables combine categorical, temporal, environmental, input, scale, and outcome information suitable for exploratory analysis.

The final dataset contains 19,685 records, 10 features, 55 crops, 30 states, and 6 seasons. The dataset was validated before EDA to ensure a consistent analytical base.

4. EDA Methodology

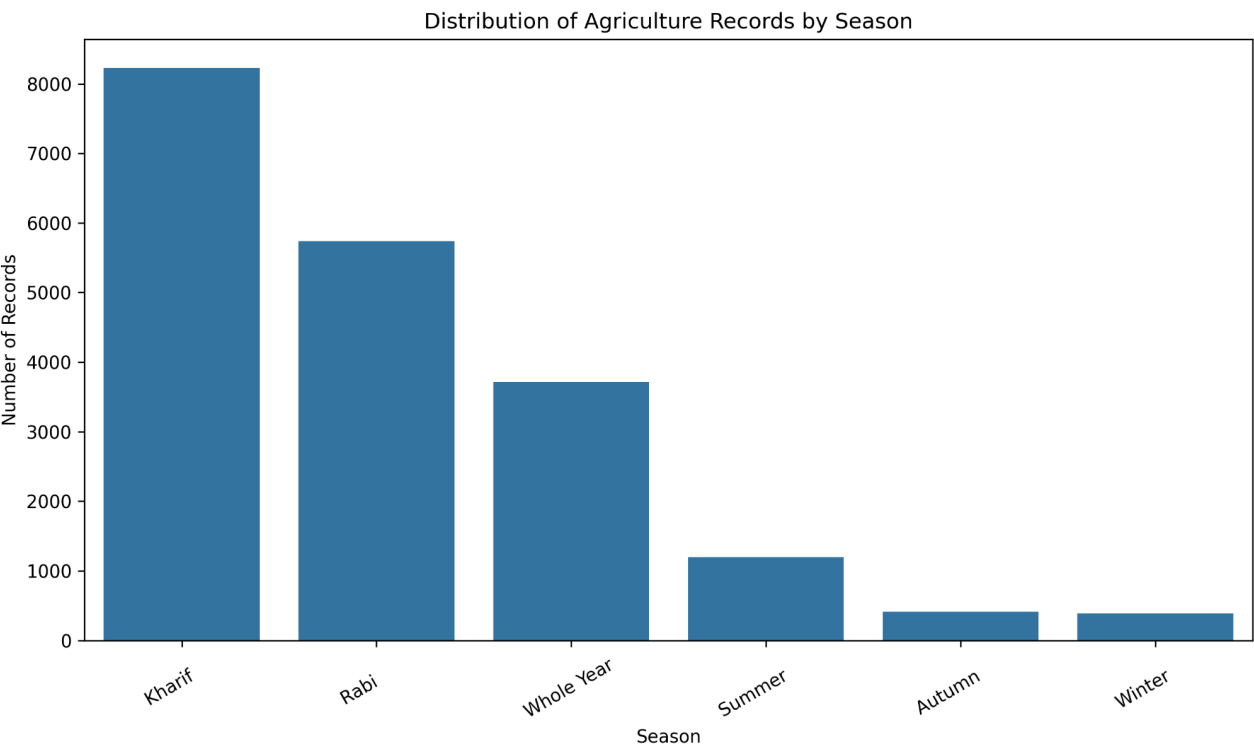
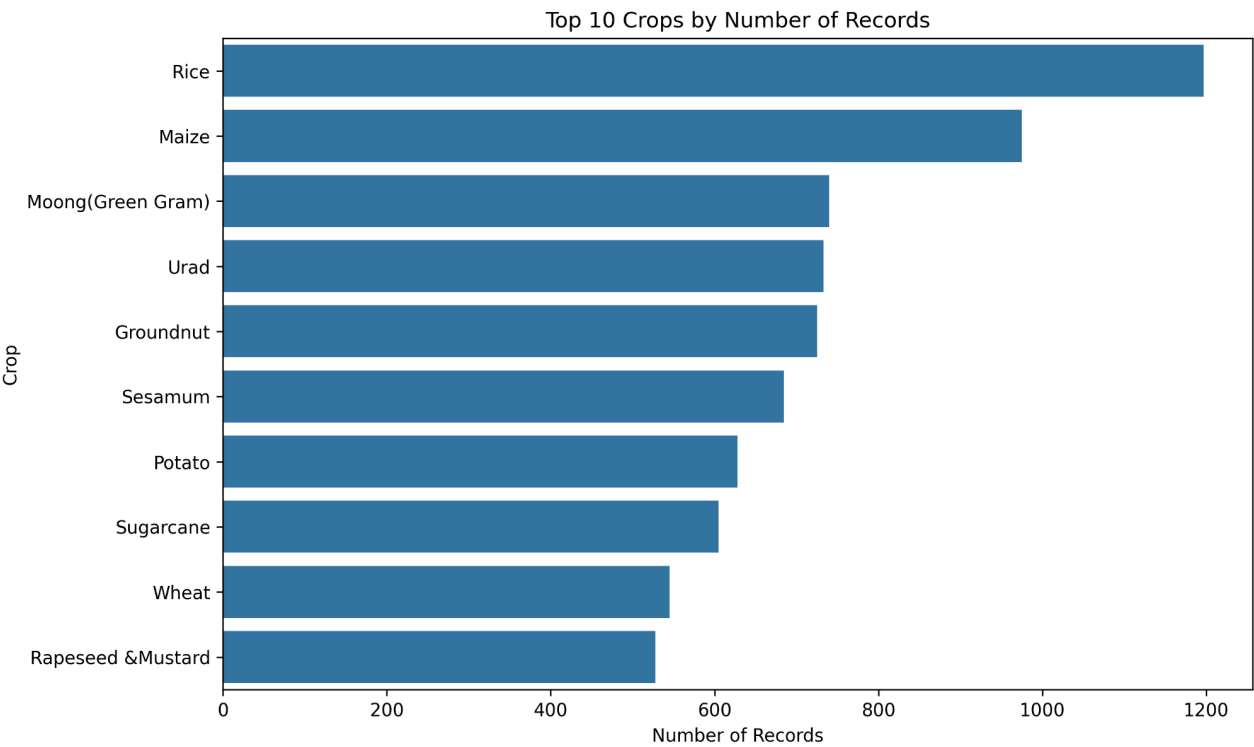
The EDA was performed through structural inspection, categorical frequency analysis, numerical descriptive statistics, distribution visualization, grouped aggregation, temporal analysis, correlation analysis, IQR-based outlier detection, and two-dimensional Crop × Season and State × Crop analysis.

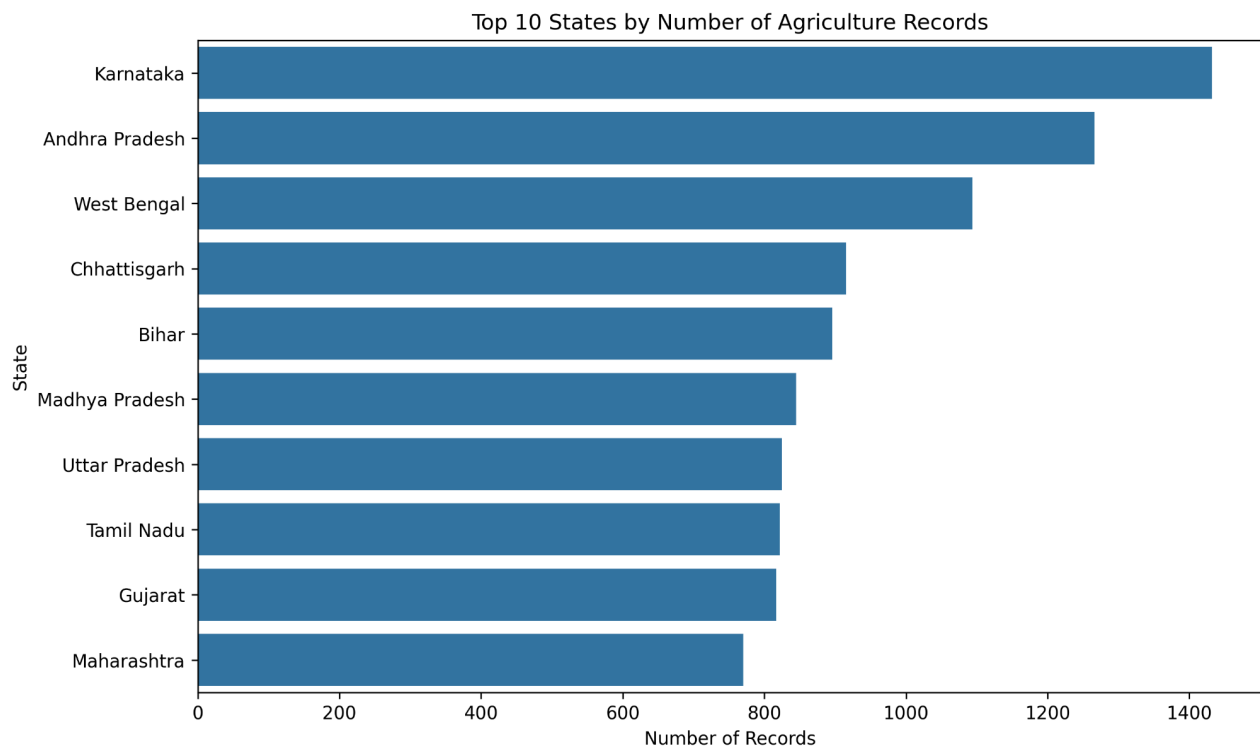
This layered workflow combines descriptive statistics with visual evidence and contextual interpretation, creating a reproducible analytical foundation for future feature engineering and predictive modeling.

5. Categorical Data Analysis

Rice is the most frequently represented crop with 1,197 records, followed by Maize with 975. Kharif is the dominant season with 8,229 records, followed by Rabi with 5,741 and Whole Year with 3,717. Karnataka has the highest record count among states with 1,432 observations.

Uneven representation is important because frequency reflects dataset structure and can affect the stability of group-level estimates and future machine learning models.



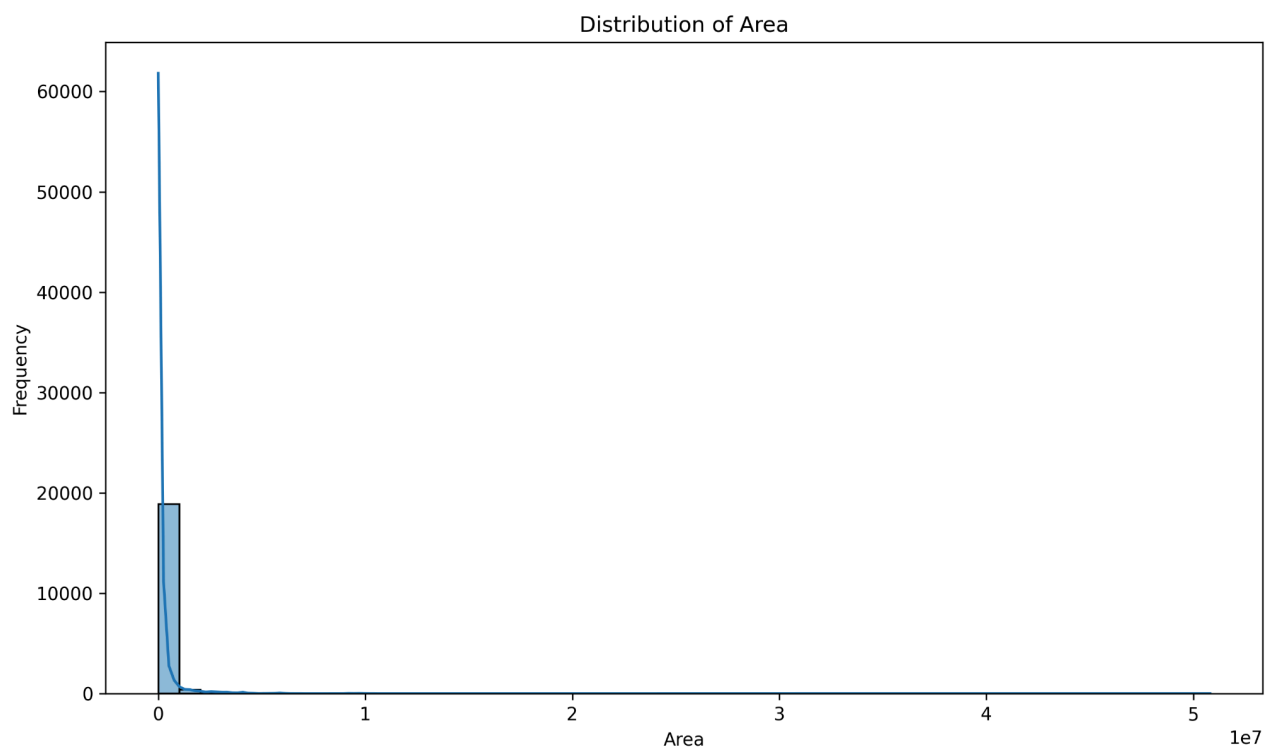
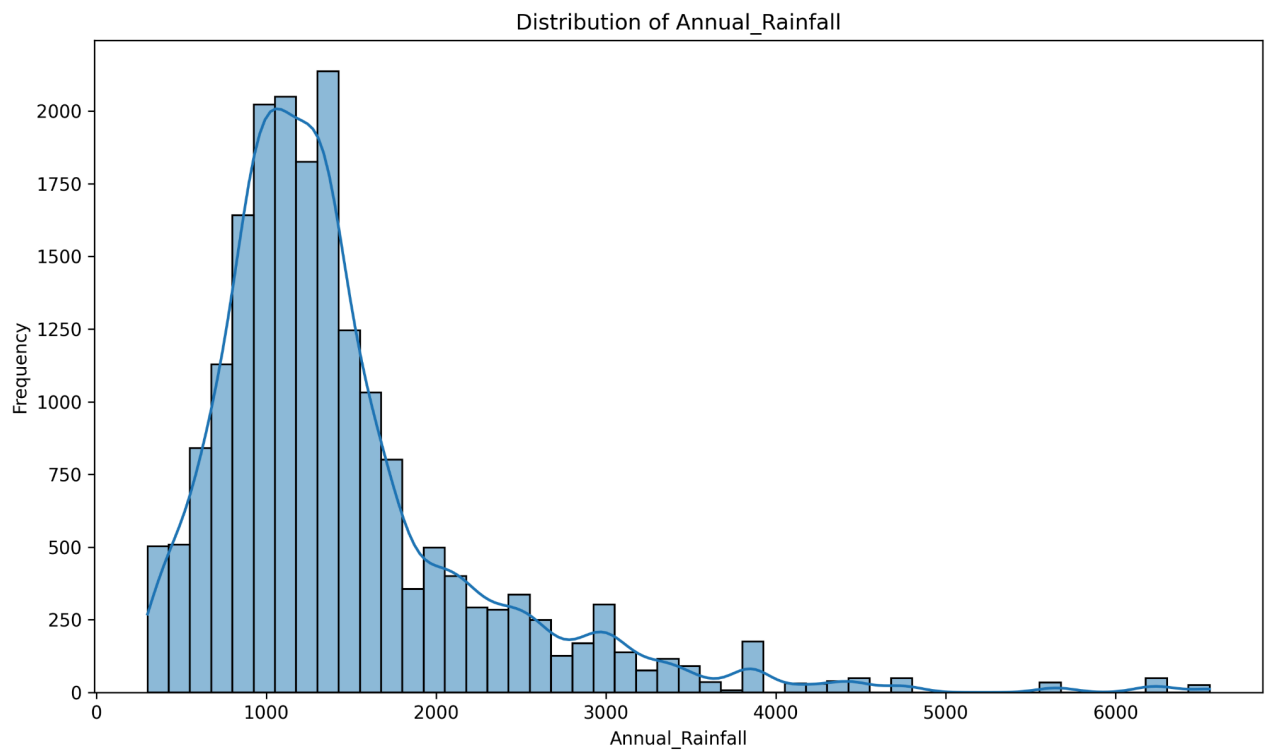


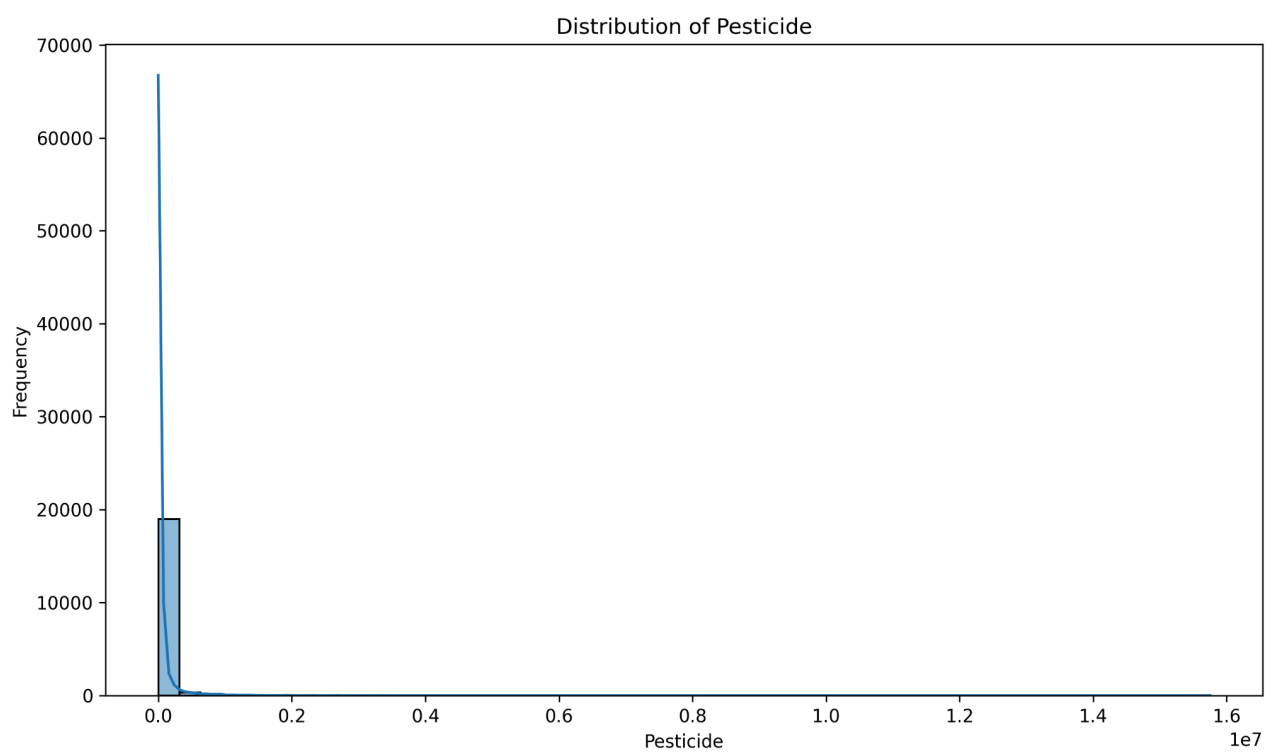
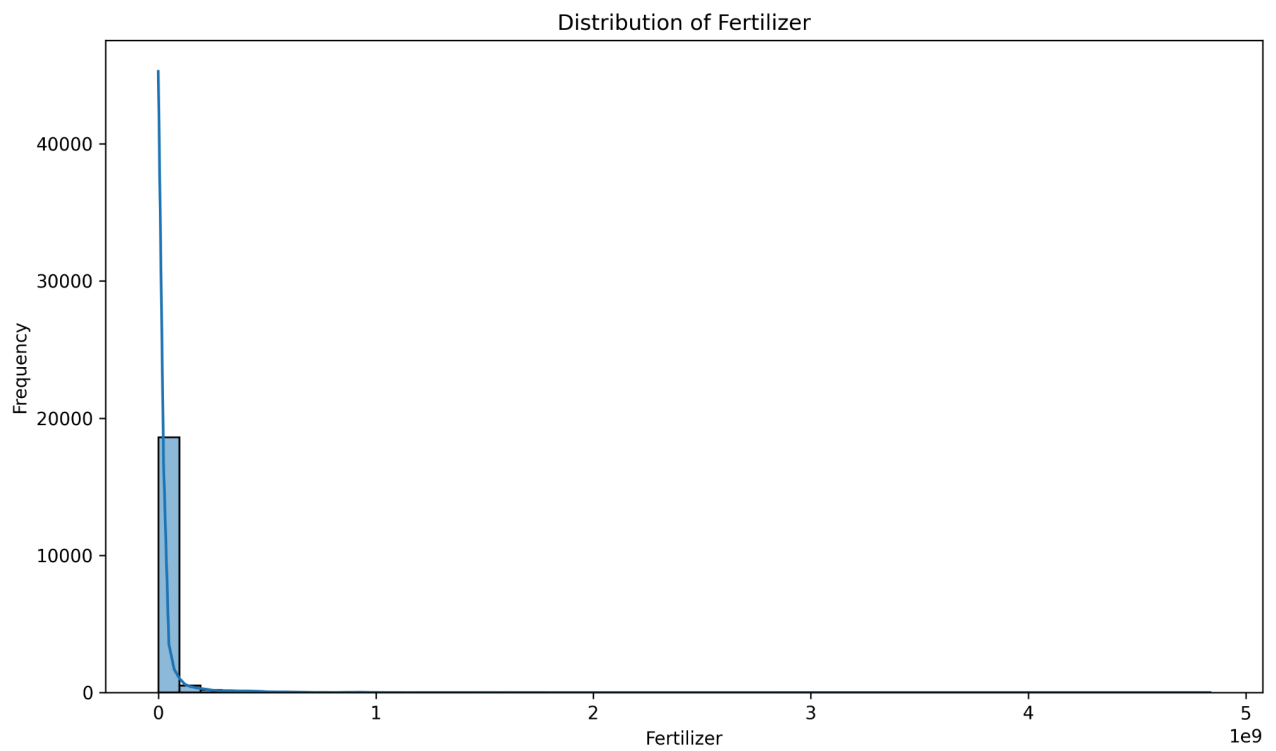
Top 10 Crops / Season Distribution / Top 10 States

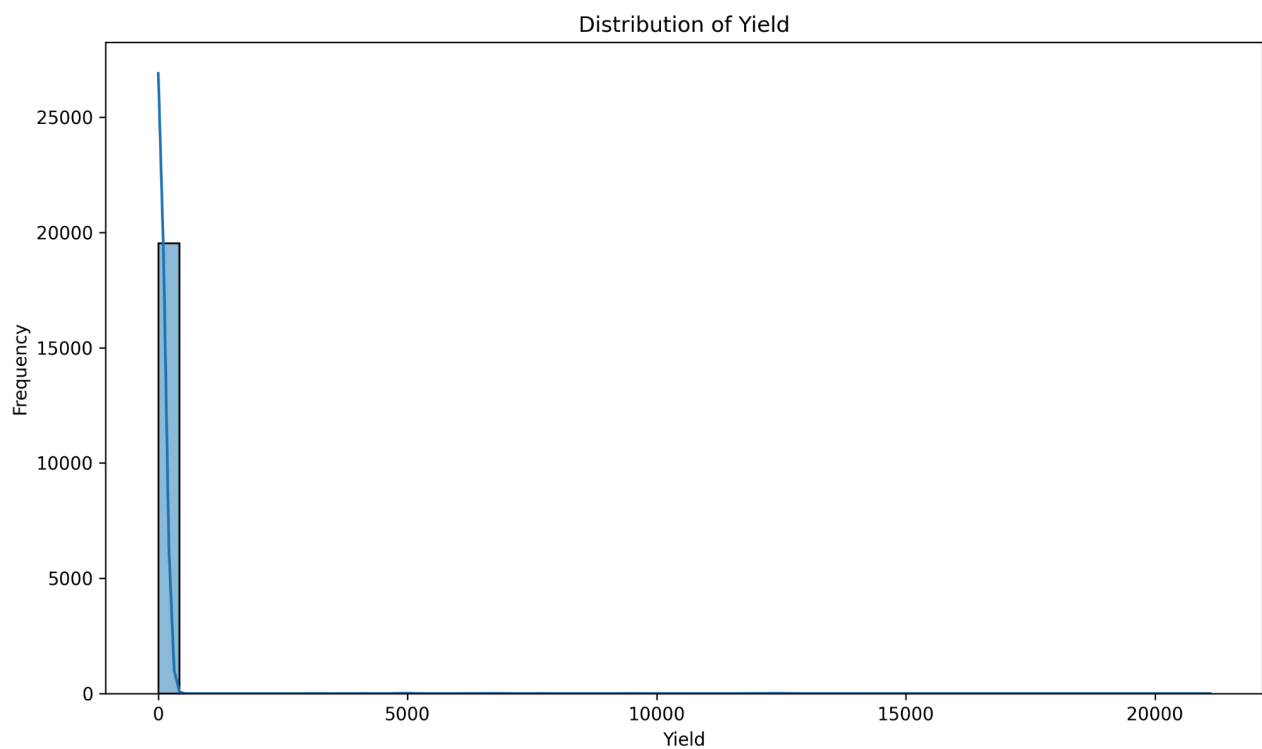
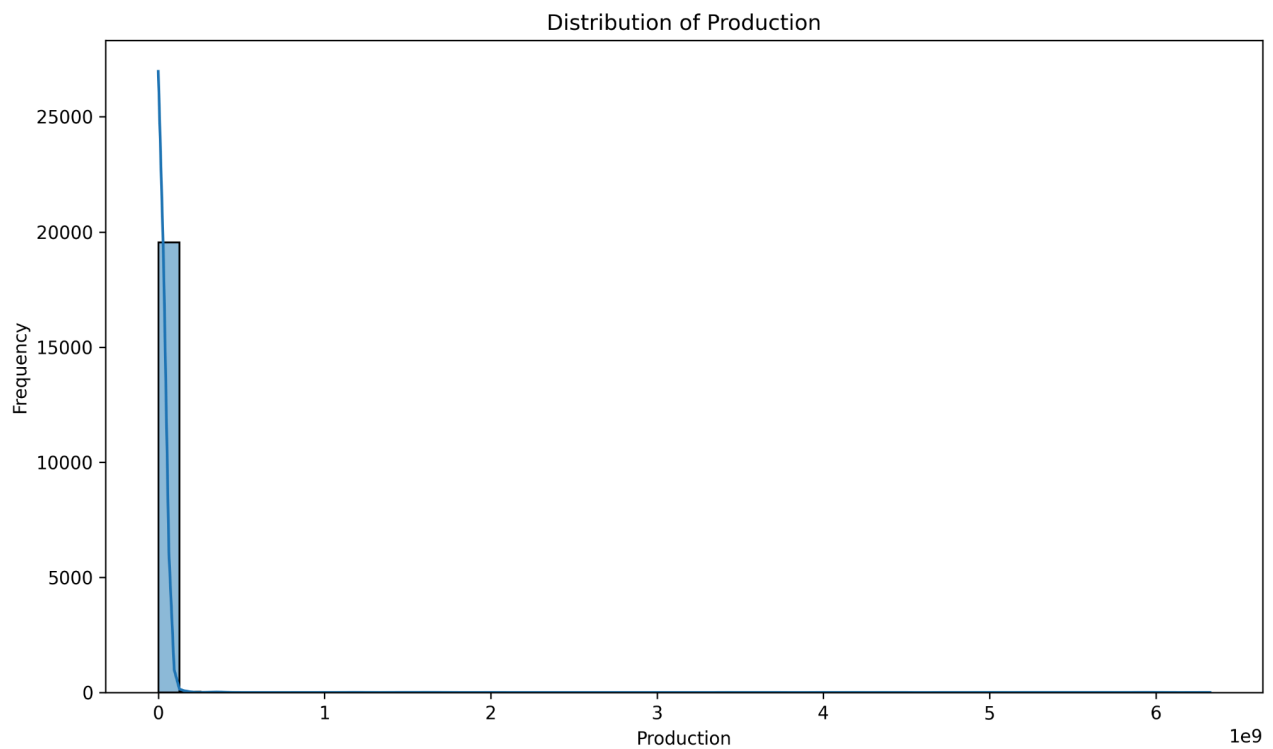
6. Numerical Distribution Analysis

Area, Production, Annual Rainfall, Fertilizer, Pesticide, and Yield were examined using descriptive statistics and distribution plots. Wide ranges and skewed distributions are expected in agricultural data because farms, crops, regions, and production scales differ substantially.

Extreme values should therefore be interpreted in agricultural context rather than treated as errors automatically. Yield is especially important because it represents productivity rather than absolute output scale.





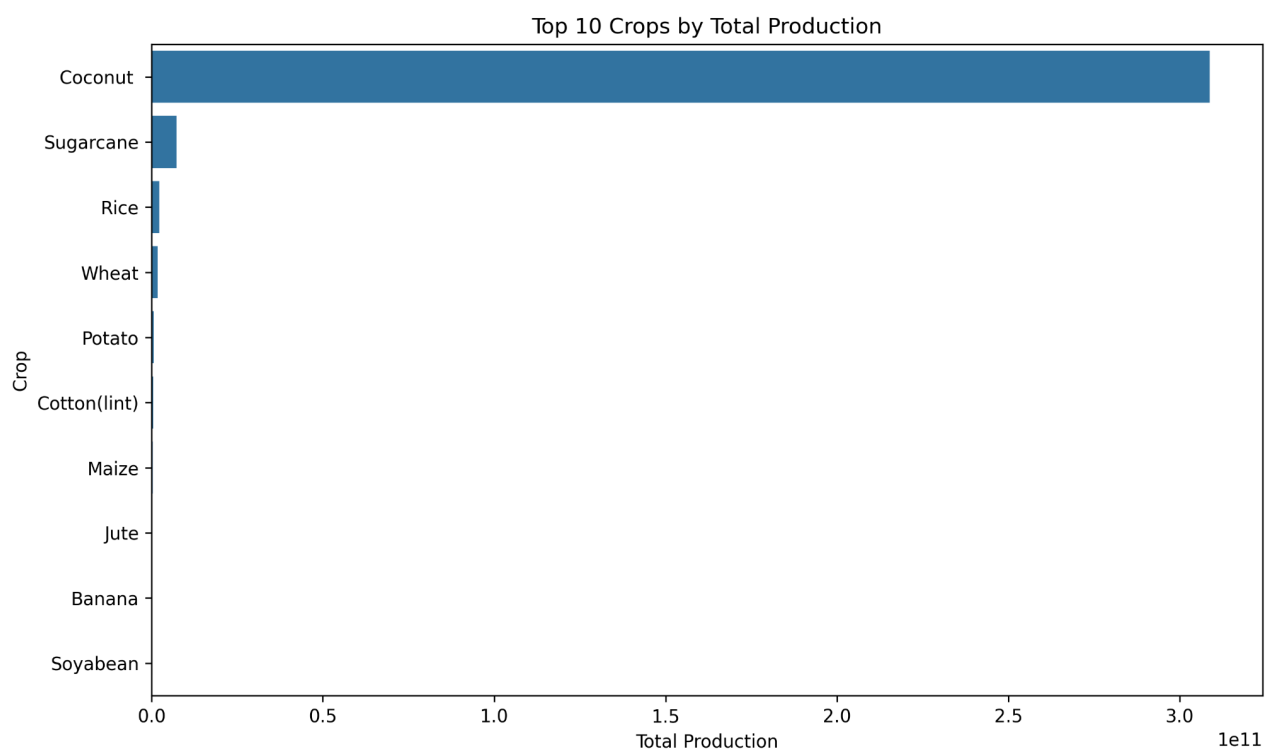
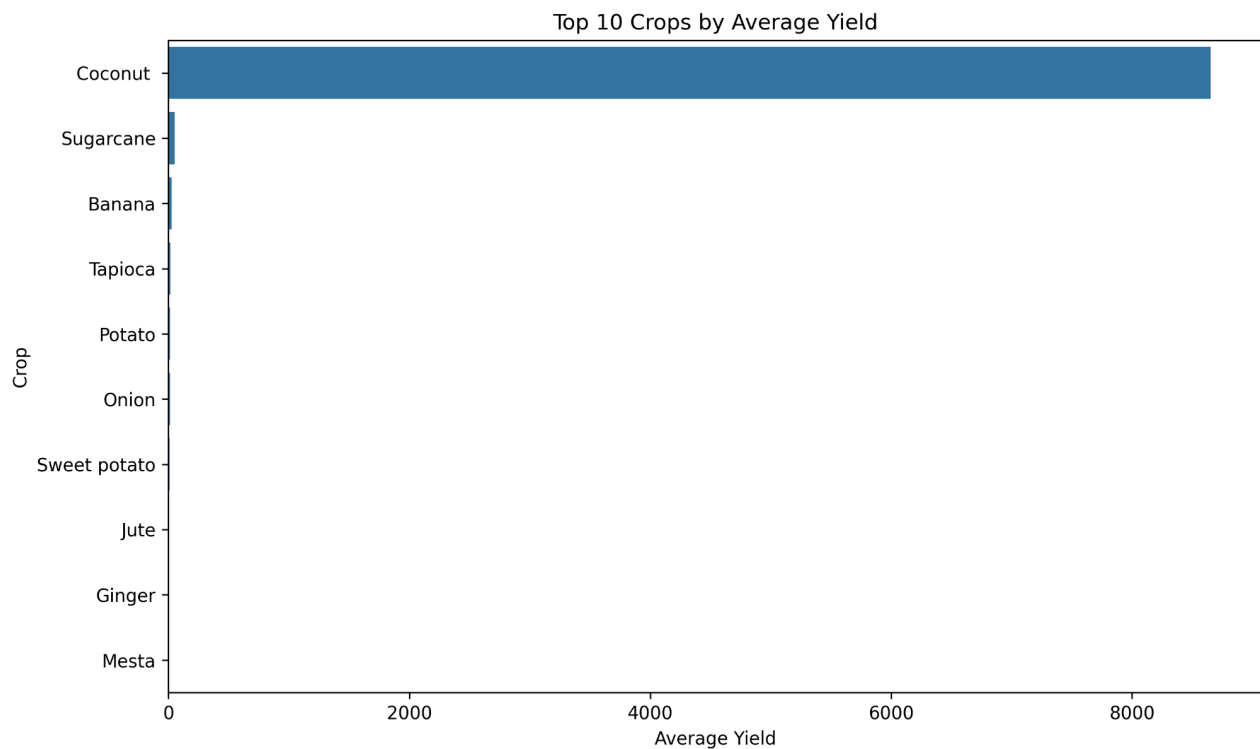


Distributions of Area, Production, Annual Rainfall, Fertilizer, Pesticide, and Yield

7. Crop-wise Agricultural Performance

Crop-wise performance was evaluated using average Yield and total Production. Coconut has the highest average Yield at 8,652.0 and also the highest total Production in the dataset, approximately 3.088 trillion in the dataset's production units.

Average Yield and total Production measure different concepts: productivity versus aggregate output. Presenting both prevents misleading conclusions based on only one metric.

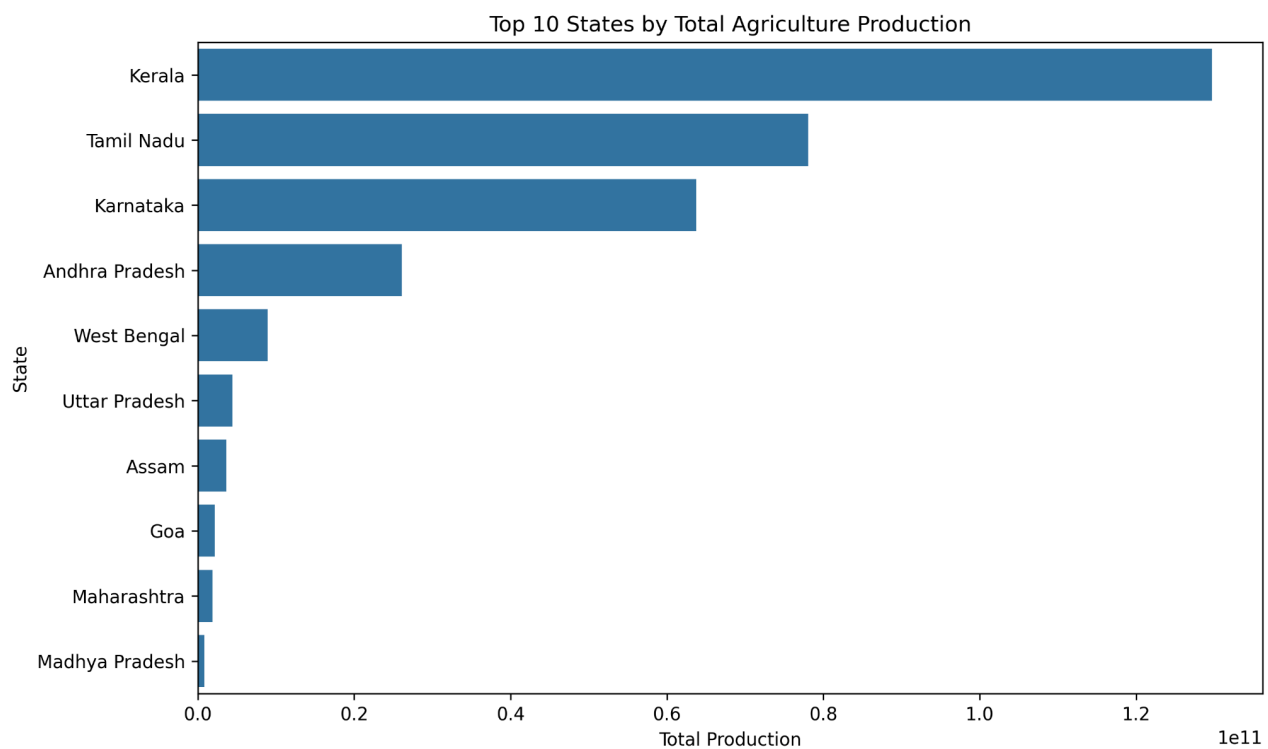
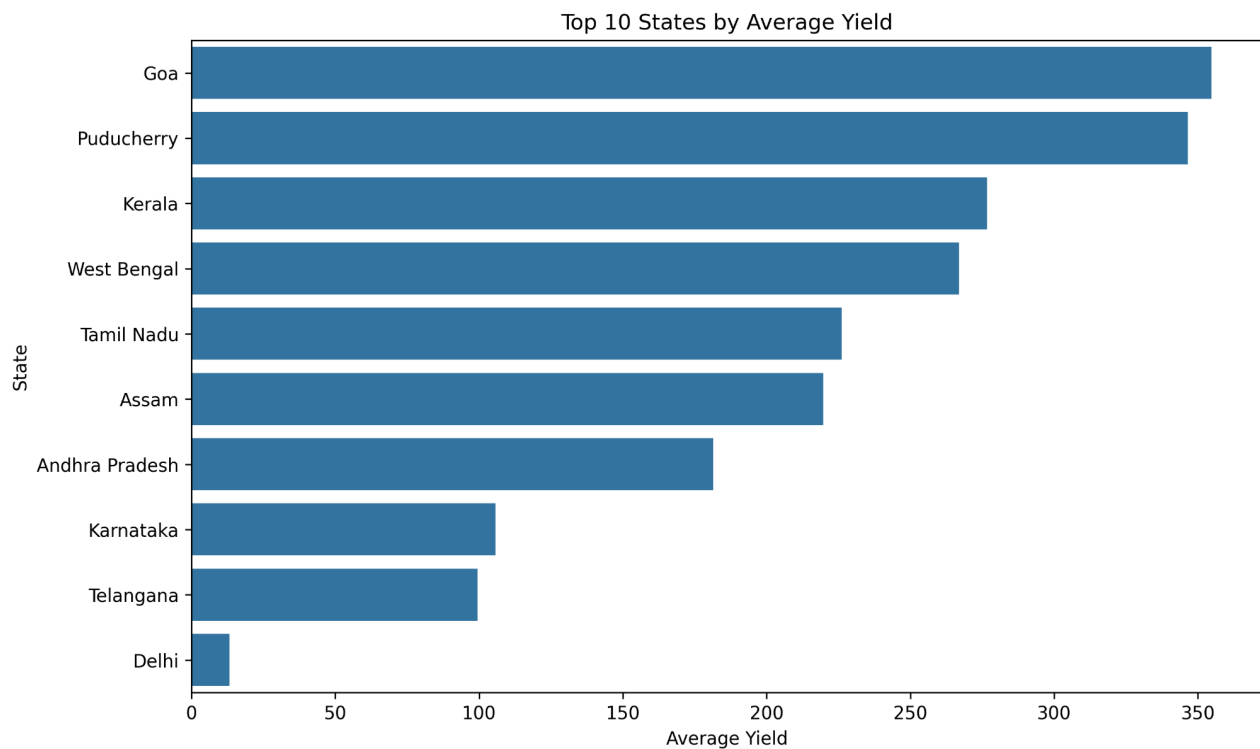


Top 10 Crops by Average Yield and Total Production

8. State-wise Agricultural Performance

Goa has the highest average Yield at 354.78, while Kerala has the highest total Production at approximately 1.297 trillion in the dataset's production units.

These results show why production and productivity should be interpreted separately. State-level performance can also reflect differences in crop mix, cultivated area, climate, soil, irrigation, and agricultural practices.

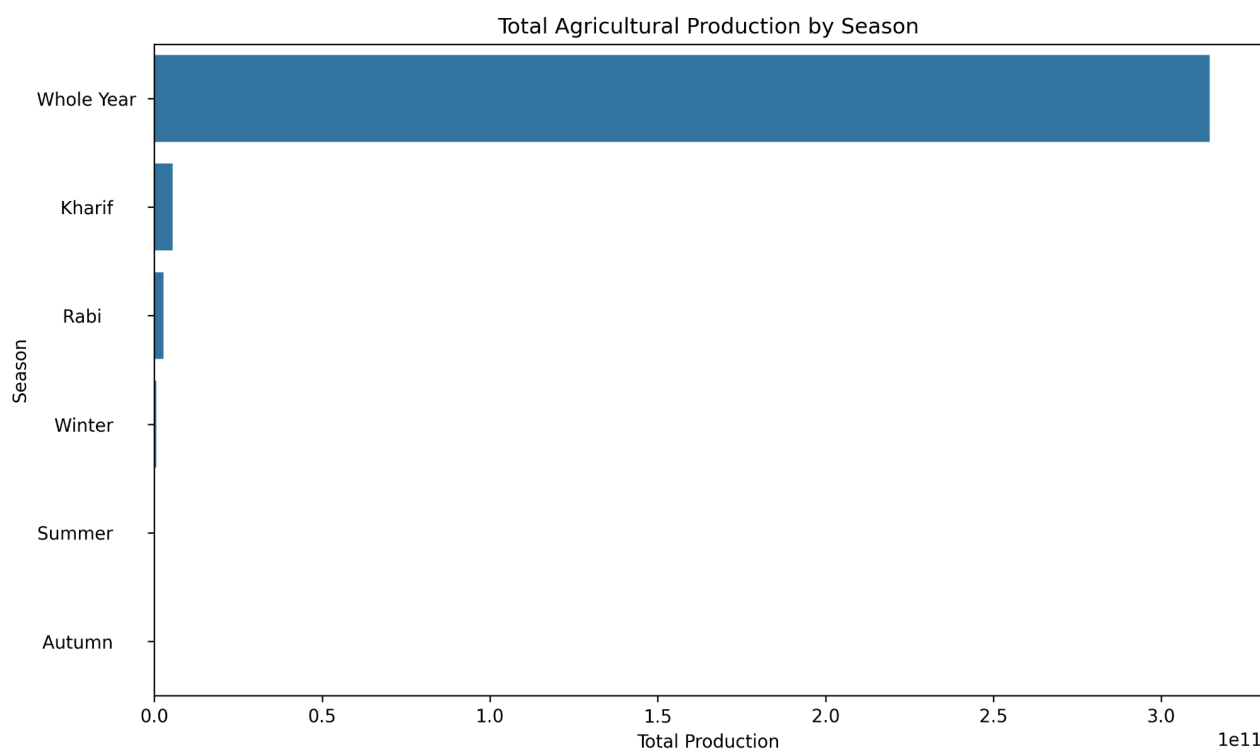
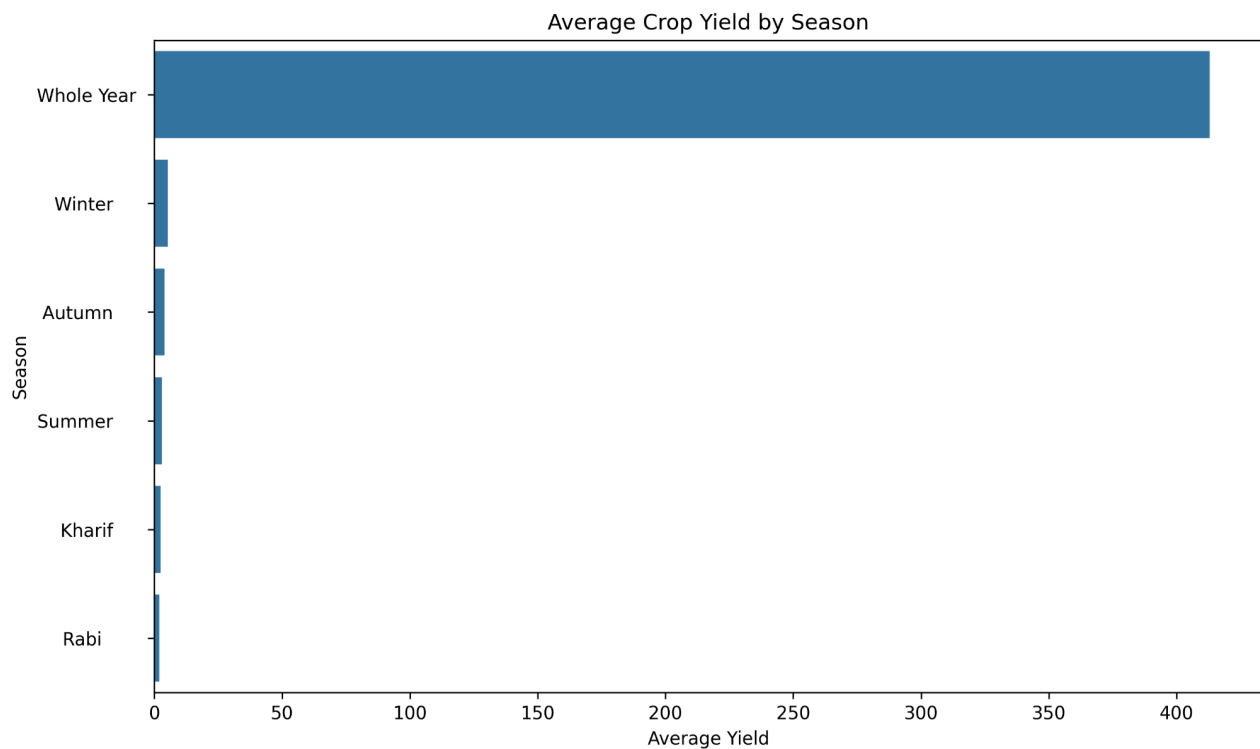


Top 10 States by Average Yield and Total Production

9. Season-wise Agricultural Analysis

Whole Year has the highest average Yield at 413.0 and the highest total Production at approximately 3.145 trillion in the dataset's production units.

Seasonal comparisons must be interpreted together with record counts because the dataset is unevenly represented across seasons.

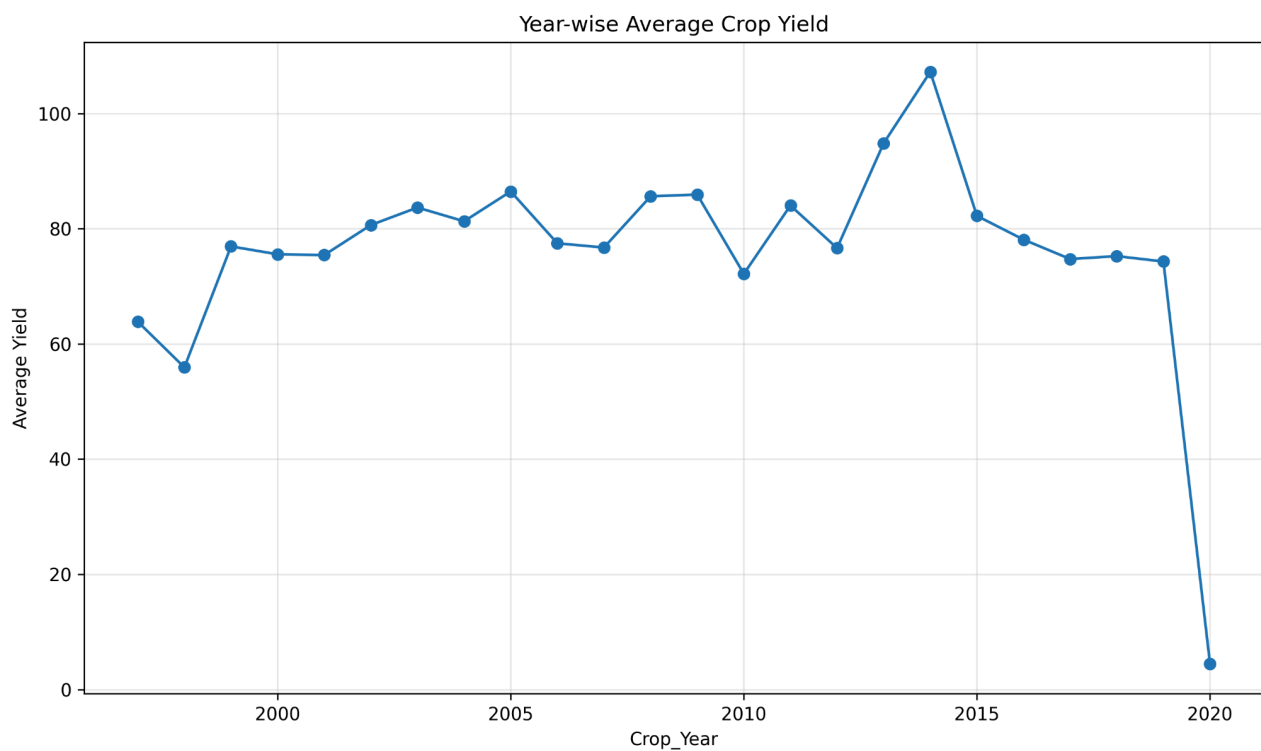
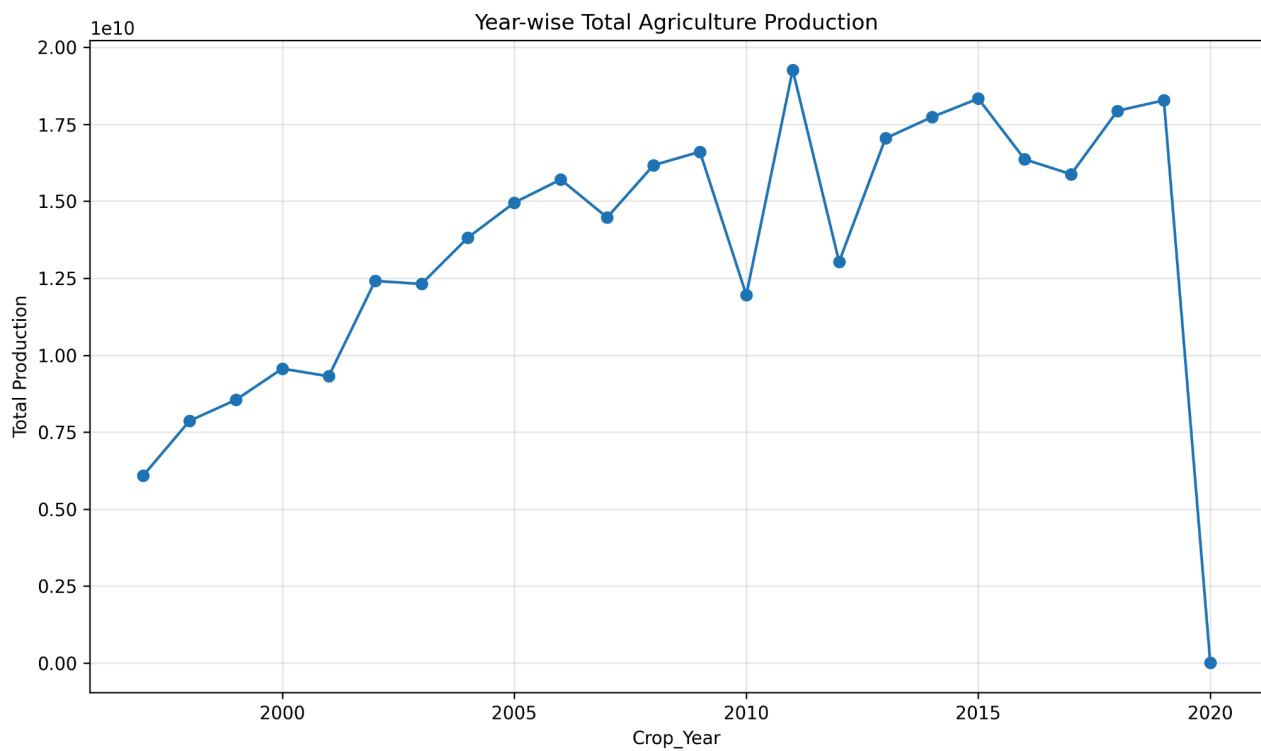


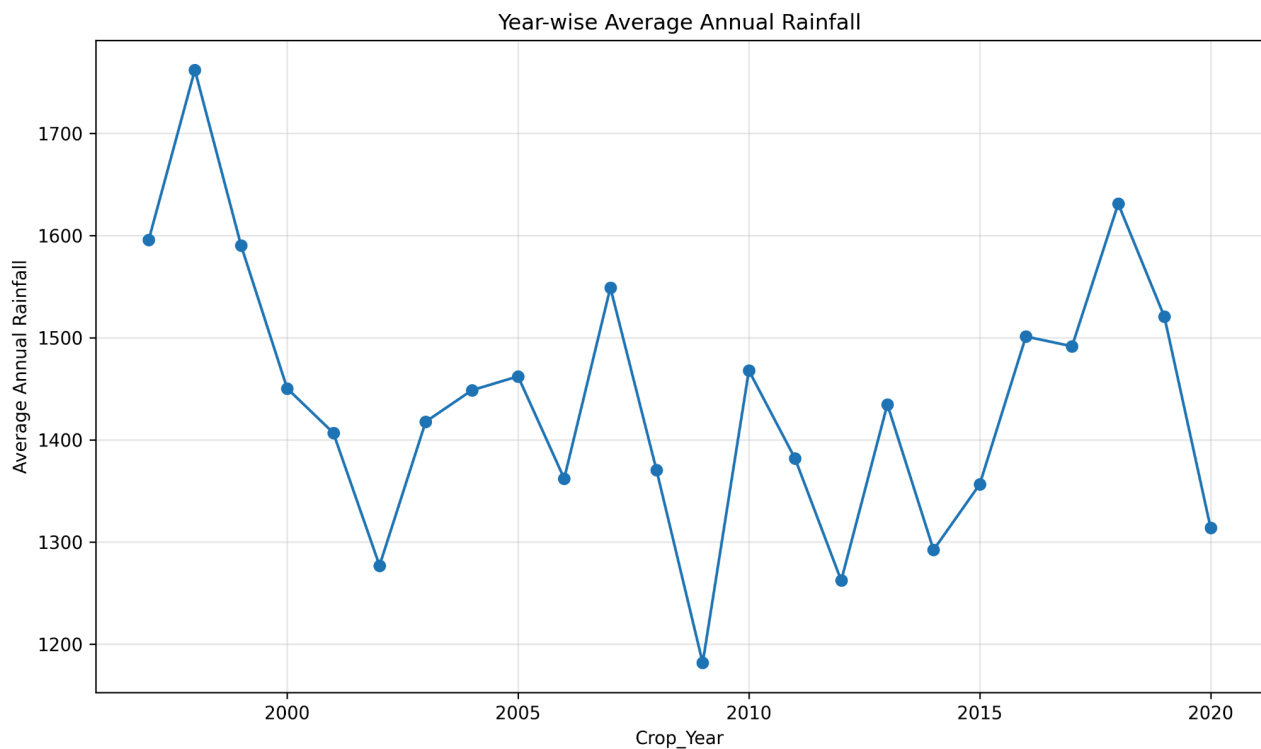
Season-wise Average Yield and Total Production

10. Year-wise Trend Analysis

Year-wise trends were analyzed for total Production, average Yield, total Area, and average Annual Rainfall. Temporal patterns can reflect changes in weather, crop composition, technology, agricultural inputs, policy, and cultivated scale.

Trend analysis identifies associations over time but does not establish causation. Production changes should therefore be interpreted alongside yield and area.



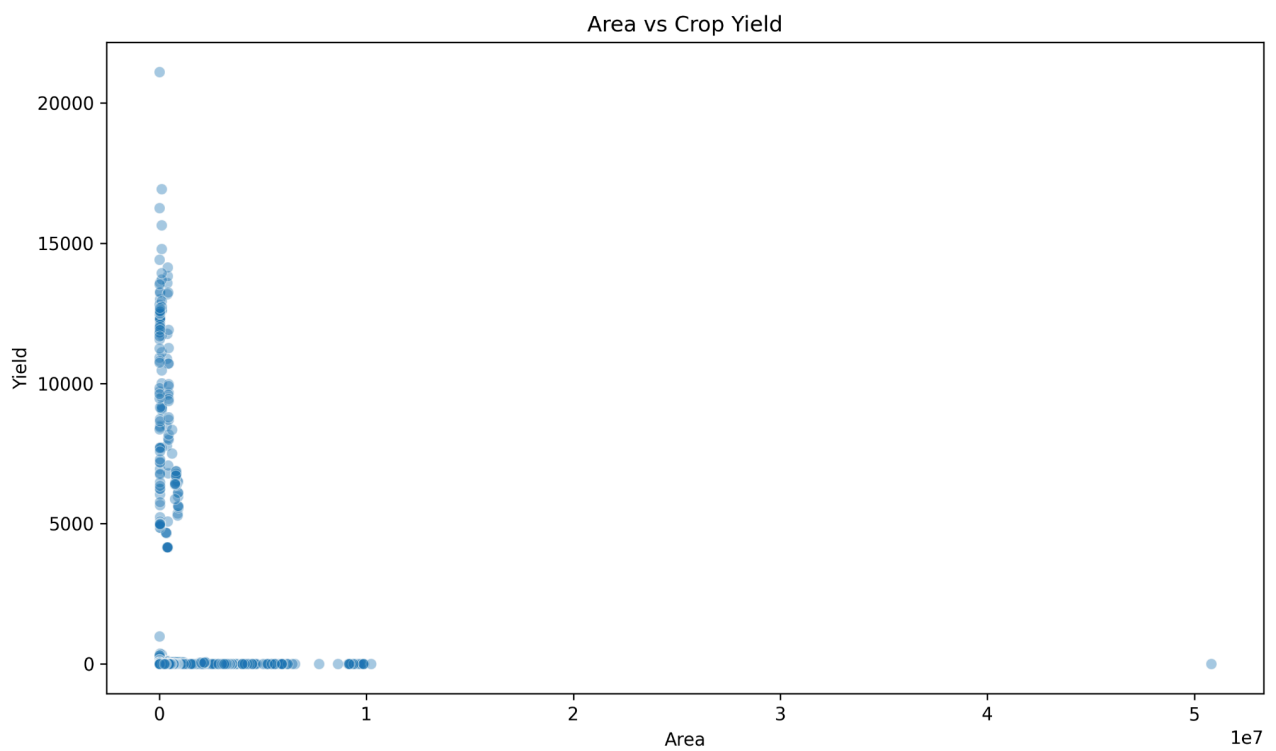
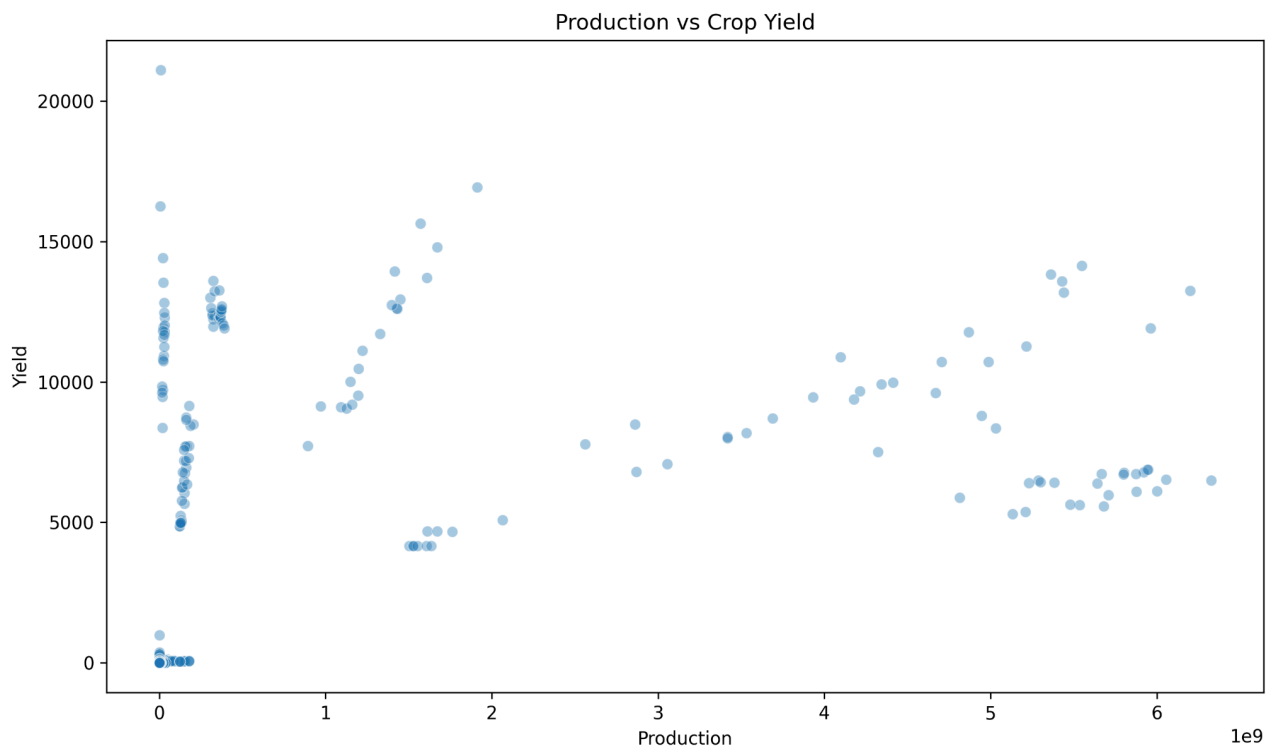


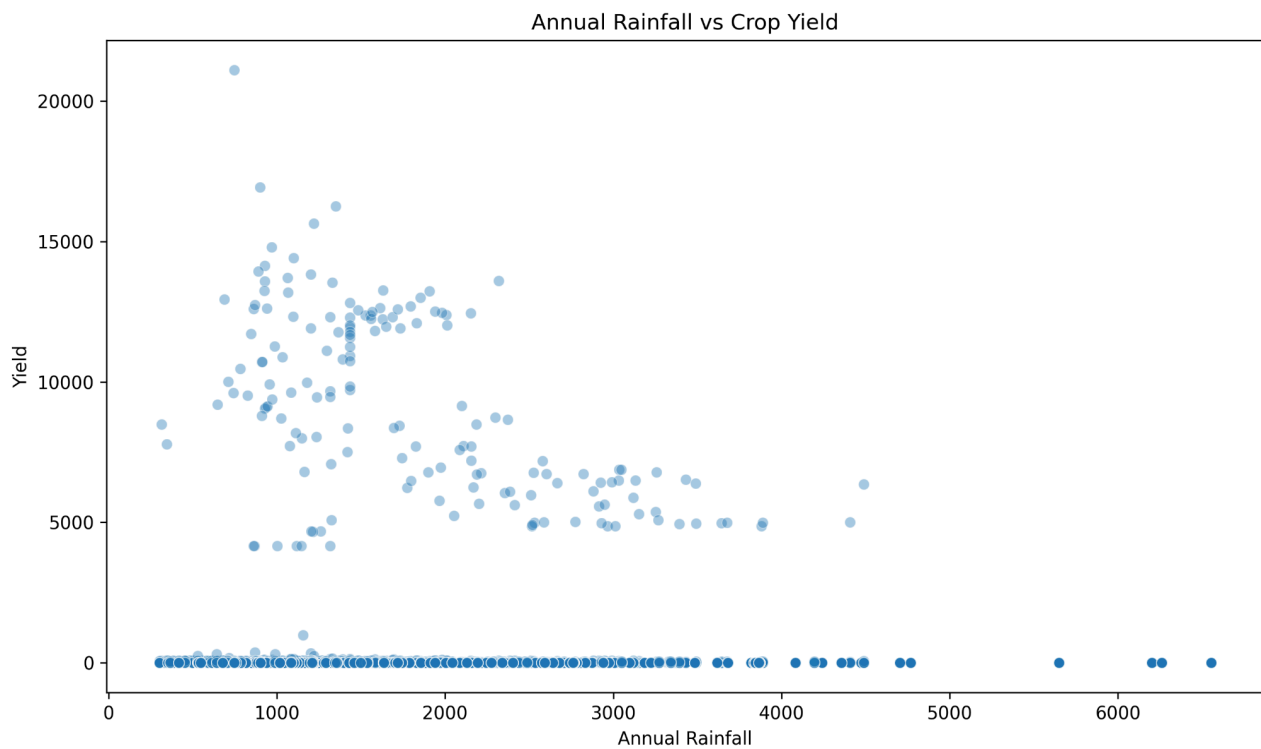
Year-wise Production / Yield / Rainfall Trends

11. Correlation and Relationship Analysis

Production has the strongest positive correlation with Yield at approximately 0.571. Annual Rainfall has a very weak positive correlation of approximately 0.021, while Fertilizer, Crop_Year, Area, and Pesticide show correlations with Yield close to zero.

Correlation is a measure of linear association and does not establish causation. Weak linear correlation also does not mean a variable is agriculturally irrelevant; effects may be nonlinear, crop-specific, regional, or interaction-dependent.



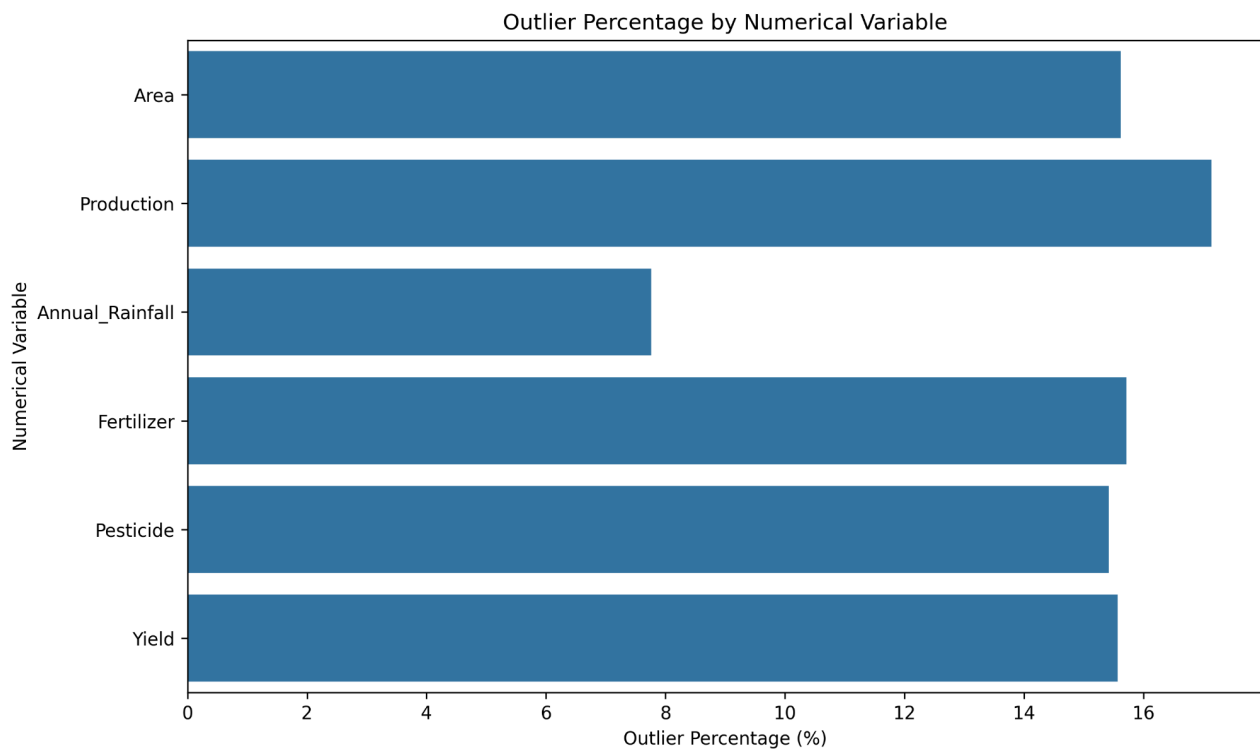


Production vs Yield / Area vs Yield / Rainfall vs Yield

12. Detailed Outlier Analysis

IQR analysis identified 3,076 Area outliers (15.62%), 3,373 Production outliers (17.13%), 1,527 Annual Rainfall outliers (7.76%), 3,093 Fertilizer outliers (15.71%), 3,036 Pesticide outliers (15.42%), and 3,065 Yield outliers (15.57%).

These observations were not automatically removed because statistical outliers may represent genuine agricultural conditions. Future modeling may consider transformations, robust scaling, or model-specific treatment.

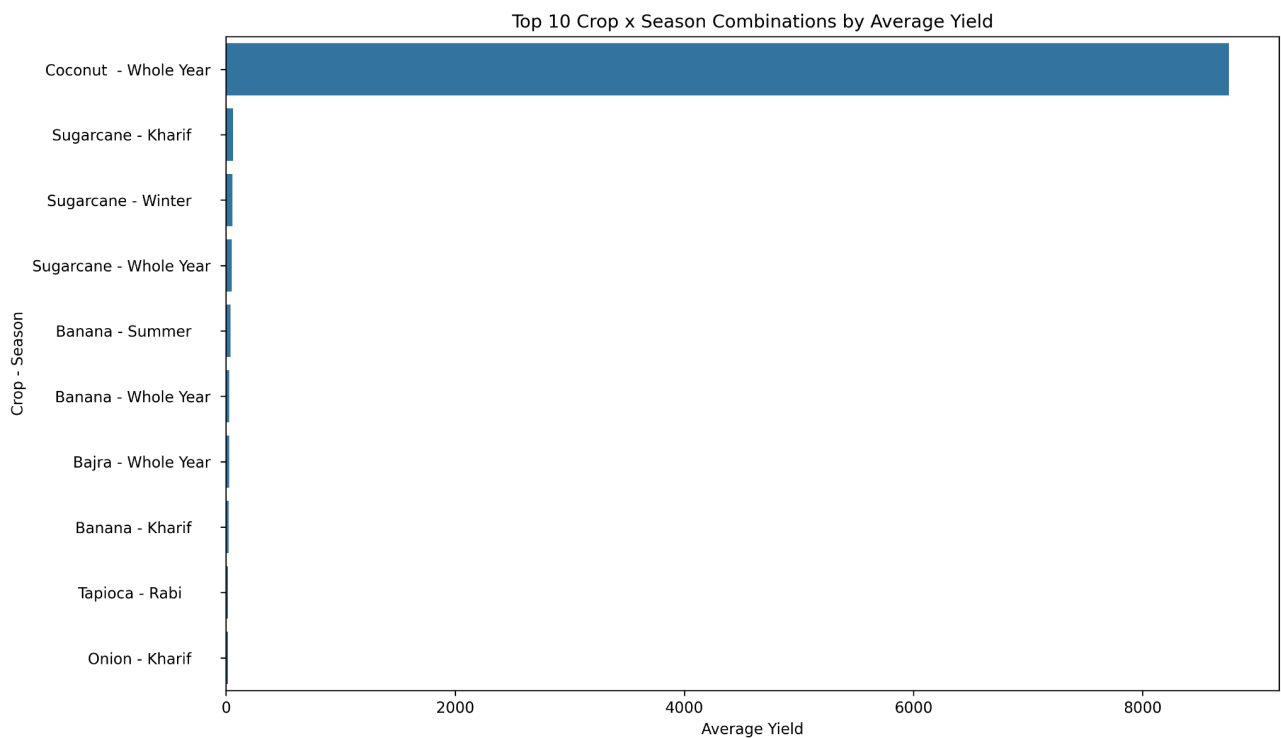


Outlier Percentage by Numerical Variable

13. Crop × Season Analysis

Crop × Season analysis combines crop and seasonal dimensions to identify changes in productivity and production across agricultural seasons. A minimum-record threshold was used for reliable ranking.

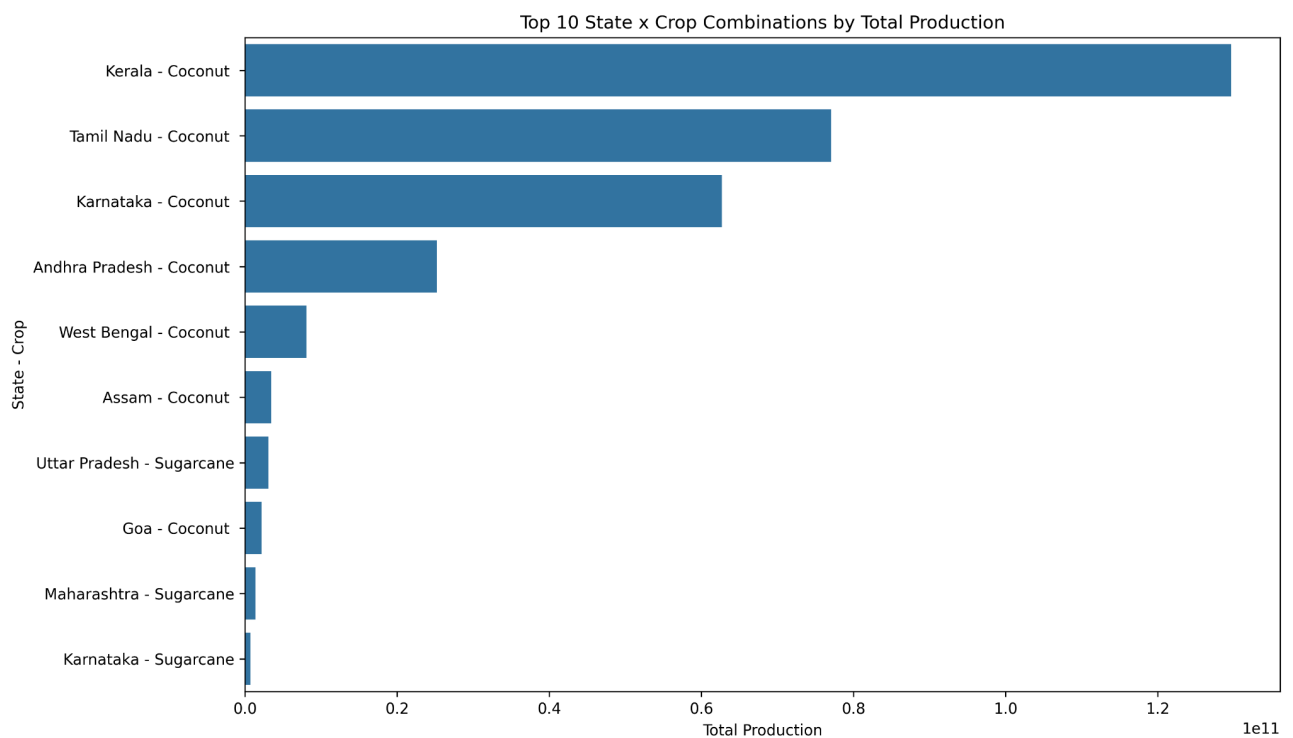
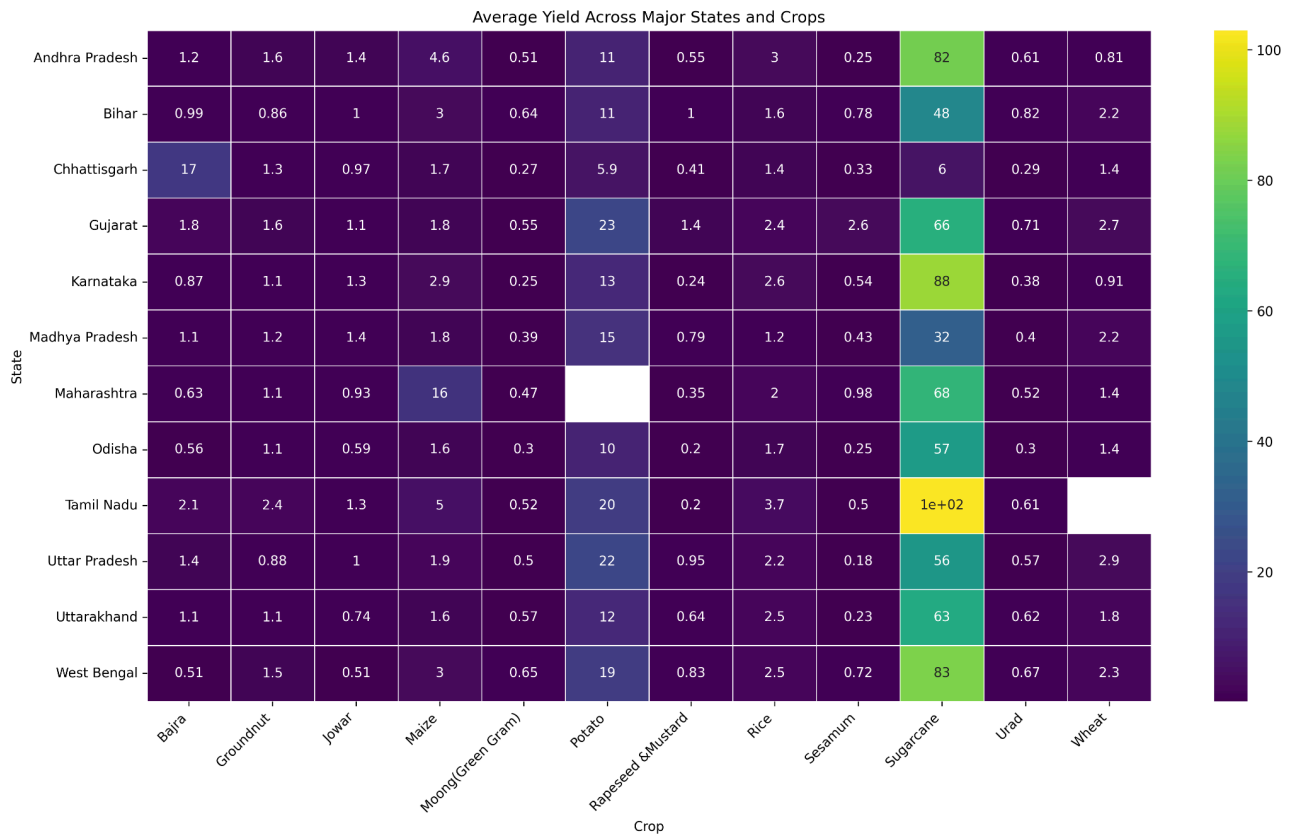
This analysis can reveal patterns hidden by one-dimensional summaries because the same crop may perform differently under different seasonal conditions.



Crop × Season Yield Heatmap / Top Combinations

14. State × Crop Analysis

State × Crop analysis investigates regional crop specialization and productivity. Combining geography and crop identity provides deeper context because agricultural performance depends on both regional conditions and crop selection.



State x Crop Yield Heatmap / Top Production Combinations

15. Data Quality Validation

The final dataset contains 19,685 records and 10 variables. Missing values, duplicate rows, and negative numerical values are all zero. There are 108 zero-production records and 112 zero-yield records, which were retained where not established as invalid.

The validation confirms that the cleaned dataset is suitable for structured statistical analysis while preserving potentially meaningful agricultural observations.

16. Key Findings and Interpretation

Rice is the most represented crop; Kharif is the most represented season; Karnataka has the most records; Coconut has the highest average Yield and total Production; Goa has the highest average state Yield; Kerala has the highest total state Production; Whole Year has the highest average Yield and total Production; Production has the strongest positive linear relationship with Yield; and all six major numerical variables contain statistical outliers.

17. Limitations and Analytical Considerations

The analysis is observational and correlation-based. It cannot establish causal relationships. Uneven representation across crops, states, and seasons can affect group-level stability. Aggregated production can also be strongly influenced by cultivated area and record composition.

Future work should evaluate category balance, nonlinear relationships, feature interactions, transformations, and appropriate validation strategies before predictive modeling.

18. Conclusion

The Week 2 EDA converts the cleaned agricultural dataset into a structured set of statistical and visual insights. The analysis identifies major differences in crop productivity, production scale, seasonal performance, and regional agricultural patterns.

The results demonstrate that agricultural performance cannot be explained by a single variable. Crop, state, season, cultivated scale, and environmental factors should be considered together. The EDA therefore provides a strong bridge from data cleaning to feature engineering and machine learning.

19. Reproducibility and Project Resources

The analysis should be reproducible through the project repository containing the cleaned dataset, analysis scripts, generated summaries, visualization files, and documentation. Before submission, the GitHub repository link should be inserted and tested.

GitHub Repository: <https://github.com/sameerkbsameer234-design/Agribusiness-ML-Data-Analysis>